

5 (**short**) An incompressible flow moves in circular, concentric streamlines between radius R_0 and R_1 , as illustrated in Fig. 2. The velocity at $r = R_0$ is V_0 .

(a) Derive an expression from first principles for the gradient of the pressure normal to the streamlines in terms of ρ , v , and r , where v is the local value of the velocity. [4]

(b) The Bernoulli constant is the same for all streamlines (i.e. it does not depend on r). By differentiating the Bernoulli equation in the direction normal to the streamlines, and using the result from (a), or otherwise, find the distribution of velocity $v(r)$, and give the value V_1 of the velocity at $r = R_1$. [6]

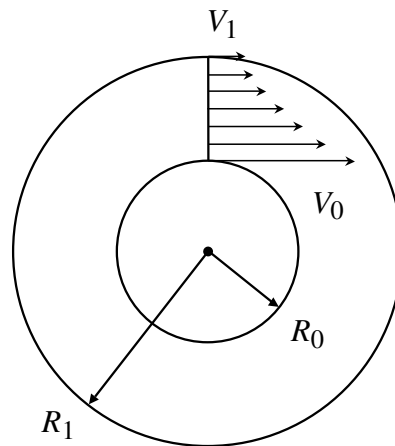


Fig. 2